Reliability Assessment of Wi-Fi CSI-Based Activity Recognition under Environmental Shift and Structured Measurement Loss

A participant-disjoint study of joint environment–cohort transfer and post-normalization observation loss.

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Abstract

Wi-Fi channel state information (CSI) supports contactless activity recognition, but clean accuracy alone does not establish reliability under domain change and incomplete measurements. We evaluate 1,260 genuine recordings from 30 participants in three indoor environments using participant-disjoint source partitions and leave-one-environment-out tests. Logistic regression and random forest attain pooled target accuracy of 36.7% and 39.0%. An initial 60-fit, five-seed comparison evaluates compact CNNs, the implemented missingness-informed model MIRA-CSI, and the official 9,996-feature MiniRocket transform. MIRA-CSI reaches 23.7% complete-input accuracy; MiniRocket reaches 53.1%. A frozen 90-fit follow-up isolates MIRA's explicit mask input, support-distance input and coverage pooling, and adds residual CNN controls. The clean-trained residual model reaches 33.8% complete-input and 26.7% contiguous-loss accuracy after averaging all five masks. Component controls do not establish an individual benefit for every MIRA addition; omitting distance slightly improves mean block-loss macro F1 in this experiment. Separately, source-only, fold-specific representation selection raises the known MiniRocket comparator to 57.4% clean accuracy. Three offline repairs are compared under synthetic loss of normalized packet positions, with reconstruction error distinguished from classification and probability quality. Checkpoint hashes and saved probabilities undergo independent checks; participant-cluster resampling preserves repeated-view dependence. Classical interaction contrasts have both signs, while clean-source calibration does not guarantee reliable confidence after transfer and loss. Different participant cohorts occupy each environment, preventing identification of a causal room effect. These results document an executable model, controlled component tests and a reproducible reliability assessment; broader performance and deployment claims require independent data.

Keywords: Wi-Fi sensing; channel state information; activity recognition; environmental shift; missing measurements; predictive reliability.

1. Introduction

Wi-Fi networks provide an existing medium for contactless sensing in homes, classrooms, and workplaces. Human motion perturbs multipath propagation, and these perturbations can be represented through channel state information (CSI). Learning-based systems can use this representation to infer activity without a camera or a body-worn device [1, 2]. Their usefulness depends on understanding how predictions behave when the sensing conditions depart from those represented in the training data.

Two sources of variation merit joint attention. The first is *environmental shift*: changes in room geometry, furniture, or sensor placement alter the propagation paths encoded in CSI [1]. The second is *structured measurement loss*, where consecutive observations are unavailable. When these conditions occur together, both recognition performance and the confidence attached to a decision may change. A plausible activity label and a high predicted probability can therefore provide an incomplete account of reliability.

Domain adaptation has been investigated as a means of reducing sensitivity to changing domains [1]. A recent review also identifies environmental robustness, consistent evaluation, and reproducibility as continuing challenges [2]. These findings motivate a controlled assessment of how measurement availability interacts with environmental shift.

2. Related work

Zinys et al. study domain-independent activity recognition using a generative adversarial network and simplified CSI preprocessing [1]. Their work addresses sensitivity to changes in sensing conditions. It provides a relevant robustness baseline, but reported accuracies cannot be transferred to a different dataset, split, or degradation model.

DensePose From WiFi maps signal phase and amplitude to dense body-surface coordinates across 24 human regions [3]. This is a spatial reconstruction

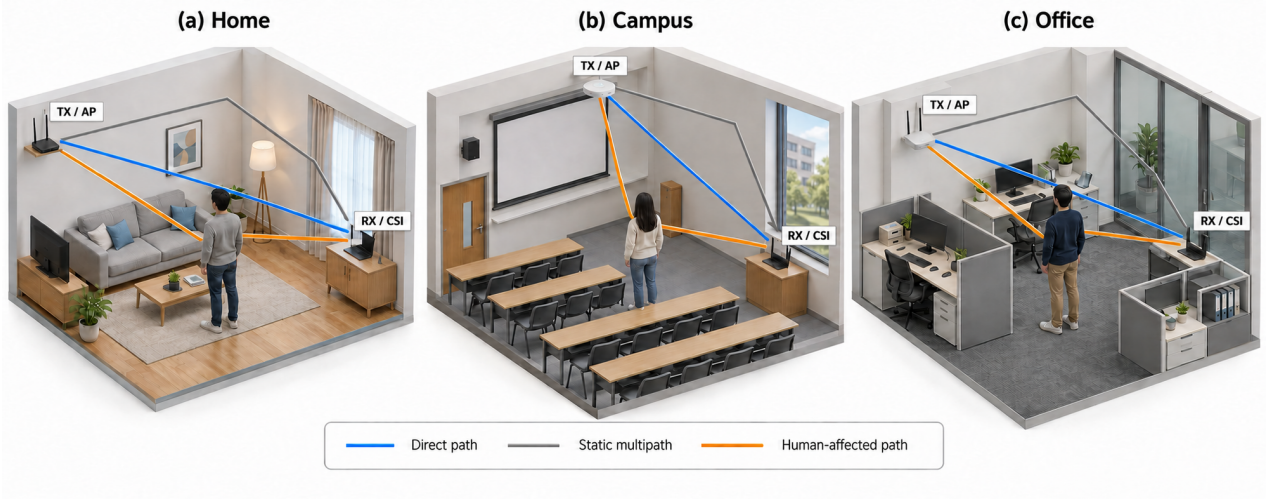


Figure 1. Illustrative home, university Campus, and office geometries. Blue identifies a selected direct path, gray selected static multipath, and amber a selected human-affected contribution. TX/AP sends and RX/CSI samples the combined channel. These explanatory scenes are not measured study sites or a ray-tracing solution.

task, whereas the present study predicts discrete activity labels. Dense pose estimates do not establish that a generic router can measure physiological signals or reproduce those results in every room. The preprint is therefore background on sensing capability, rather than a directly comparable activity classifier.

Widar3.0 estimates body-coordinate velocity profiles to reduce dependence on environment, position, and orientation [4]. Its central insight is that a physically informed representation can suppress domain-specific signal variation. This motivates evaluating robustness across explicit domain partitions, while distinguishing gesture recognition from whole-body activity recognition.

SenseFi provides a common implementation and benchmark for several Wi-Fi sensing datasets and learning architectures [5]. The authors describe UT-HAR as a single-environment dataset whose sliding-window samples can overlap. They also describe NTU-Fi HAR recordings from three layouts. These distinctions matter: random sample splits and environment-held-out splits answer different questions. Dataset identity alone does not establish that a processed release preserves the metadata needed for an independent cross-environment evaluation.

Learned handling of incomplete CSI predates the present implementation. CSI-BERT reconstructs missing CSI using self-supervised bidirectional representations [18]; CSI-BERT2 extends pretraining and fine-tuning to CSI prediction and classification with temporal embeddings and adaptive subcarrier re-weighting [19]. Zakeri et al. encode the age of retained samples relative to the session decision and fuse it with CSI embeddings [20]. These approaches establish prior work on missingness and temporal context. Their reported performance is not a matched baseline here, because datasets, acquisition assumptions, training access and tasks differ.

Guarino et al. examine the availability of datasets, preprocessing descriptions, and code needed to reproduce CSI sensing studies [2]. Their analysis motivates retaining file-level provenance, domain identifiers, and complete evaluation settings. We document the releases actually used and audit their sampling units before interpreting performance.

Guo et al. distinguish classification accuracy from probability calibration and evaluate post-hoc temperature scaling [6]. Their empirical findings concern the evaluated neural networks and datasets; they do not establish that calibration will survive CSI degradation. Here, calibration parameters are fitted using source validation observations only and frozen before test evaluation.

Geifman and El-Yaniv frame selective classification as a trade-off between prediction risk and coverage [7]. Following this distinction, a model that accepts fewer observations must be compared at a matched acceptance rate. We compute full-coverage recognition alongside confidence-ranked selective error; rejection alone is not interpreted as improved recognition. Distribution shift also requires separate assessment; a risk guarantee proved under specific assumptions will not be imported into this evaluation without checking those assumptions.

The experiment combines an environment factor with a measurement-availability factor. Random masks and contiguous missing blocks are compared at matched missing fractions using the same test recordings and frozen models. Masks are controlled corruptions of genuine recordings, not observed network outages.

Alsaify et al. provide recordings with explicit environment, subject, experiment, activity, and trial

identifiers [8]. Their author repository also supplies parsed MAT files and amplitude-scaling code [9]. These labels permit environment exclusion and participant-disjoint partitions. The three environments contain different participant cohorts, which limits causal interpretation of the environmental comparison.

Our contribution is a reproducible reliability evaluation of this combined transfer and measurement-loss setting. It includes recognition, probability quality, class errors, and conditional participant-grouped uncertainty. It is an evaluation of fixed accessible baselines, rather than a claim of a new pose-reconstruction architecture or a best-performing classifier.

3. Propagation and measurement

Wi-Fi sensing observes a communication channel. A transmitter emits radio signals and a compatible receiver estimates how propagation changes their amplitude and phase across subcarriers and antenna links. This channel state information is more detailed than a single received-power indicator. Halperin et al. describe a practical interface for collecting such measurements from a supported wireless card [10]. The channel estimate is the input to an inference procedure; it does not itself contain an anatomical image.

An explanatory multipath model for subcarrier k and antenna link r is

$$\widehat{H}_{k,r}(t) = \sum_{p=1}^{P} a_{p,r}(t) e^{-j2\pi f_k \tau_{p,r}(t)} + \eta_{k,r}(t), \tag{1}$$

where $a_{p,r}$ is the complex coefficient of path p, $\tau_{p,r}$ is its delay, and $\eta_{k,r}$ collects noise and residual measurement effects. This simplified representation explains superposition, rather than solving the electromagnetic field in the illustrated rooms. Practical CSI also contains hardware timing and phase effects that must be considered during preprocessing [2].

The received signal combines relatively direct contributions with paths affected by walls, furniture, and people. Body movement can change path length, attenuation, and scattering. Their complex sum can increase or decrease even when the person's movement is smooth. For a change in a path's propagation length ΔL_p, the corresponding phase change is approximately

$$\Delta\varphi_p = -2\pi \Delta L_p / \lambda, \tag{2}$$

where λ is wavelength. This sensitivity to geometry motivates temporal and physically informed features, including the body-coordinate velocity profiles studied in Widar3.0 [4].

For a single moving scatterer and stationary transmitter and receiver, differentiating the path phase gives the instantaneous Doppler contribution

$$f_{D,p} = \frac{1}{2\pi}\frac{d\varphi_p}{dt} = -\frac{1}{\lambda}\frac{dL_p}{dt} = -\frac{\mathbf{v}\cdot(\mathbf{u}_T + \mathbf{u}_R)}{\lambda}. \tag{3}$$

Here $\mathbf{v}$ is scatterer velocity, and $\mathbf{u}_T$ and $\mathbf{u}_R$ point from the transmitter and receiver, respectively, toward that scatterer. This is a bistatic single-bounce geometric relation; multiple body segments contribute different path changes. It explains motion sensitivity, but does not provide a unique inversion from CSI to human pose. The present experiment uses amplitudes only and does not estimate Doppler or recover calibrated phase.

The colors in Figures 1 and 2 have explicit explanatory roles. Blue identifies a selected transmitted or direct contribution, amber identifies a human-affected contribution, and gray identifies selected reflections involving static surroundings. They do not encode measured strength, distinct frequency bands, or recognition confidence. The person scatters and attenuates the incident field; the amber contribution does not imply that the person actively emits a Wi-Fi signal.

The arcs schematize cross-sections of wavefronts, meaning equal-phase locations of an idealized contribution. They are drawn widely apart for legibility. Radio propagation fills a volume, and the receiver samples the superposed field rather than a sequence of narrow rays scanning anatomical landmarks. A standing torso, bent seated limbs, and a horizontal body change the geometry differently. However, different combinations of position, furniture, and posture can produce similar observations. A reliable classifier must be tested on the variations it is intended to handle.

Water content provides physiological context for body–wave interaction. The often quoted 60% refers to an illustrative whole-body mass fraction, not a universal constant or a measured value for the participants in this dataset. The USGS describes adult body water as up to about 60% and notes variation with age and body composition [11]. Water is distributed within tissues rather than pooled in a separate compartment shaped like part of the silhouette.

For radio propagation, the relevant material description is frequency-dependent dielectric response. Water polarization and mobile ions contribute to the response of heterogeneous biological tissues, whose properties differ by tissue type. Gabriel and Gabriel report tissue measurements and dispersion modelling over broad RF and microwave frequency ranges [12]. An effective complex relative permittivity can be written, using the $e^{j\omega t}$ convention, as

$$\epsilon_r^*(\omega) = \epsilon_r'(\omega) - j\frac{\sigma_{\mathrm{eff}}(\omega)}{\omega\epsilon_0}, \tag{4}$$

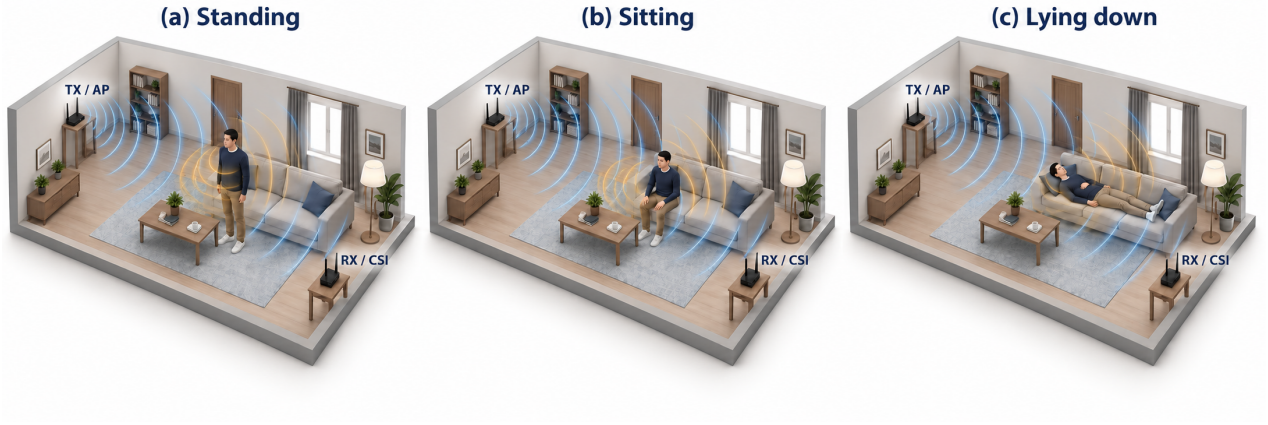


Figure 2. Fixed-room comparison of standing, sitting, and lying. Blue arcs represent illustrative incident wavefronts; amber arcs represent illustrative body-scattered contributions. Arc spacing and magnitude are not to scale or computed by a wave solver. These illustrative postures explain geometry and are not classes evaluated by the six-activity pilot.

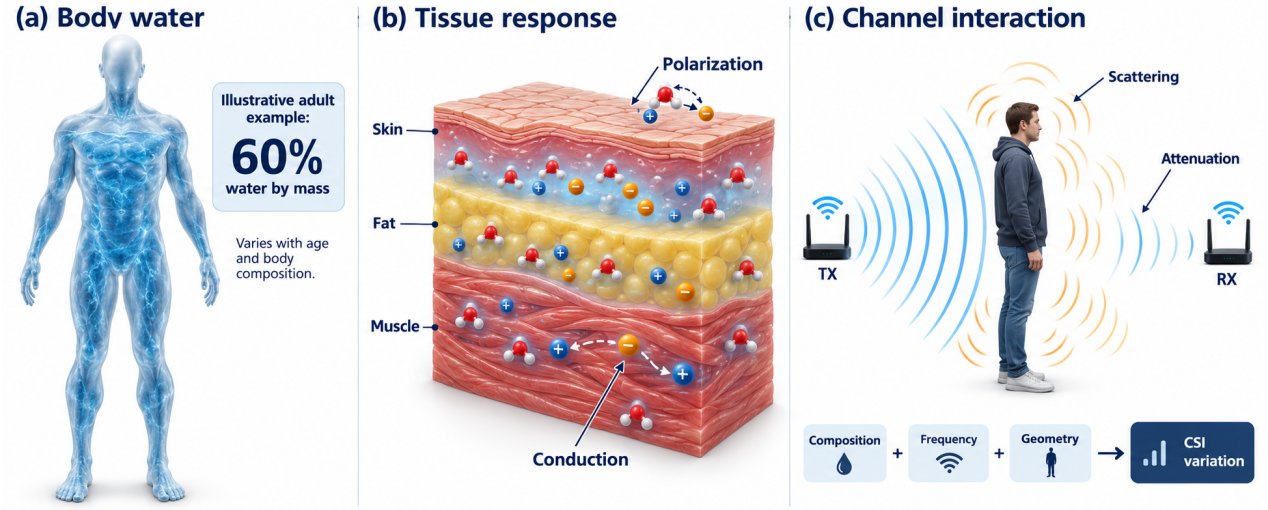


Figure 3. Body water, tissue response, and channel interaction. The 60% badge is a qualified illustrative adult mass fraction [11], not a measurement of a study participant. Tissue layers, molecules, ions, and wavefronts are schematic and not to scale; icon counts do not encode concentrations. Water and ions contribute to dielectric response, while frequency and geometry also determine propagation [12]. This explanatory figure does not estimate hydration or depict a measured RF field.

where ϵ_r' describes the real part and σ_{eff} combines effective loss mechanisms in this representation. These quantities influence phase and attenuation; interfaces, incidence direction, polarization, and body geometry also affect scattering. The whole-body water fraction cannot be substituted for a tissue permittivity or converted directly into an activity label.

Figure 3 therefore links composition to a material response and then to propagation. Neither experiment measures body water or tissue dielectric properties. Observed CSI variations cannot be interpreted as hydration estimates, temperatures, or subject-specific water percentages.

The inference pipeline separates radio observations from an estimated activity or rendered human representation. A receiver first exports CSI observations. Preprocessing aligns the time series, handles missing samples, and constructs input features. A trained model then estimates an output defined by its supervision. The present study returns one of seven activity labels and a probability vector. DensePose From WiFi instead estimates body-surface correspondence [3]. A green wireframe body is therefore a representation of a different inference target, not a physical shape formed by radio waves in the room.

Deployment context changes the inference problem. In a home, moving the sofa or the access point changes background propagation. A campus classroom adds different dimensions, desk arrangements, and potentially several moving occupants. In an office, partitions and occupancy patterns introduce further channel variation. These factors motivate separate environment and occupancy evaluations

rather than transferring a single-room score to other buildings [2].

The word *Campus* identifies a university context or a possible network name. An SSID does not specify a CSI export interface or enable sensing by itself. Acquisition needs a compatible radio, suitable firmware or driver access, packets to observe, and a defined receiver configuration. Espressif's documentation, for example, specifies explicit CSI configuration and a reception callback on supported ESP32 platforms [13]. Such support establishes how to acquire measurements. Recognition of human activity still requires a trained model, appropriate labels, and validation in the intended environment.

4. Experimental design

This study tests how activity recognition and predictive confidence change under environment-and-cohort transfer together with structured measurement loss. Familiar-environment tests also exclude training participants, so the comparison does not rely on recognizing people observed during fitting.

Table 1. Evaluated conditions by domain.

CSI input	Familiar	Held-out
Complete	Reference	Transfer
Random loss	Loss only	Joint
Contiguous loss	Loss only	Joint

Colour denotes test conditions, not accuracy.

The completed evaluation has three parts:

1. quantify separate and combined degradation;
2. compare recognition and probability quality; and
3. retain reproducible splits, settings, and predictions.

5. Mathematical formulation

Let $X \in \mathbb{R}^{C\times 500}$ denote a fixed-length representation of one recording. The measured channel has $C = 90$ scaled log-amplitude coefficients, reduced to 18 five-subcarrier bands for the implemented learning pipelines. A binary temporal mask m_t is shared by every channel: $m_t = 1$ denotes an observed time position and $m_t = 0$ a missing one. Its missing fraction is

$$q = 1 - \frac{1}{500}\sum_{t=1}^{500} m_t. \tag{5}$$

Missing entries are treated as unavailable, rather than as physical zero-amplitude measurements. For observed indices $a < t < b$ with no observed position between them, the reconstruction is

$$\widetilde{X}_{c,t} = X_{c,a} + \frac{t-a}{b-a}(X_{c,b} - X_{c,a}). \tag{6}$$

Observed entries are retained and exterior gaps use the nearest observed boundary value. This deterministic operator uses no activity label. Nevertheless, it changes signal variation and therefore the feature distribution. Random and contiguous masks are compared at exactly the same q, so missing fraction and temporal arrangement remain distinct factors.

Let $p_{i,k}$ be a baseline probability for class k on file i. The implemented temperature transformation is

$$\widetilde{p}_{i,k}(T) = \frac{\exp(\log(\max(p_{i,k},\varepsilon))/T)}{\sum_{\ell=1}^{K}\exp(\log(\max(p_{i,\ell},\varepsilon))/T)}, \tag{7}$$

where $K = 7$, $\varepsilon = 10^{-12}$, and T minimizes validation negative log-likelihood over $[0.1, 10]$. This probability transformation follows the temperature-scaling principle [6]; it does not imply that the forest has neural-network logits. The model and T remain fixed for every test condition. Positive T does not reverse class-probability order; the initial clipping can introduce ties among very small probabilities.

For n evaluated predictions, the two proper scoring rules used here are

$$\begin{aligned} \mathrm{NLL} &= -\frac{1}{n}\sum_{i=1}^{n}\log \widetilde{p}_{i,y_i}, \\ \mathrm{Brier} &= \frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K}(\widetilde{p}_{i,k} - \mathbb{1}\{y_i = k\})^2. \end{aligned} \tag{8}$$

Here y_i is the true activity and $\mathbb{1}$ is an indicator. The Brier convention is a sum over classes, without dividing by K. Numerical NLL evaluation clips probabilities at the floating-point bounds used by the scoring library; the transfer analysis uses float64 precision. For the ten confidence bins B_b, the reported calibration summary is

$$\mathrm{ECE}_{10} = \sum_{b:\,|B_b|>0}\frac{|B_b|}{n}|\,\mathrm{acc}(B_b) - \mathrm{conf}(B_b)|. \tag{9}$$

Bin accuracy is the fraction of correct labels and bin confidence is the mean maximum probability. Empty bins contribute zero. ECE depends on binning and is reported beside NLL and Brier, rather than used as the sole reliability criterion.

For confidence-ranked acceptance of a files, let S_a contain the a highest-confidence observations. Selective coverage and error are

$$C_a = \frac{a}{n}, \qquad R_a = \frac{1}{a}\sum_{i\in S_a}\mathbb{1}\{\hat{y}_i \neq y_i\}. \tag{10}$$

This operationalizes the risk–coverage distinction [7]. In each transfer fold, $a = 134$ of 168 familiar files and $a = 336$ of 420 held-out files are accepted for the saved selective-error summaries. Ranking separately within a test condition is descriptive; choosing a

deployable threshold would require independent validation.

For a fixed mask pattern, the transfer experiment uses a descriptive interaction contrast. With error $E_{e,q}$ in domain condition $e \in \{0, 1\}$ and missing fraction q, define

$$I_q = (E_{1,q} - E_{1,0}) - (E_{0,q} - E_{0,0}). \tag{11}$$

Zero denotes additive effects on this error-rate scale, and positive I_q denotes a larger loss-associated increase in the held-out condition. Here $e = 0$ denotes familiar-environment testing and $e = 1$ the excluded target environment. Because participant cohorts also change, I_q is a descriptive error contrast rather than a causal environmental interaction. It is computed separately in each fold.

6. Cross-environment data and participant partitions

The transfer dataset of Alsaify et al. contains three documented environments and 30 participants [8, 9]. E1 and E2 use line-of-sight arrangements, whereas E3 uses a non-line-of-sight arrangement. The author repository organizes parsed recordings by environment and explicitly encoded subject, experiment, activity, and trial. We use its MAT release at commit `5337147c5156`; the full revision and each Git blob hash are retained in the manifest.

Seven canonical activity segments were fixed before model scoring (Table 2). These distinguish sitting, lying, and standing from four movements. Falling means the supplied falling-from-sitting segment. It does not combine all possible fall mechanisms. Other source segments, including turning and standing up, are outside this selected task.

Table 2. Transfer-study labels and canonical source segments. C/A are source experiment/activity identifiers.

Key	Activity	C/A
SIT	Sitting still	C1/A1
FAL	Fall from sitting	C1/A2
LIE	Lying down	C1/A3
STD	Standing still	C2/A4
WLK	Walk TX to RX	C3/A6
SDN	Sitting down	C4/A11
PCK	Pick up a pen	C5/A12

Within every participant and experiment, a fixed SHA256 ranking selected six of twenty trial IDs. Related segments of the same experiment use the same selected trial IDs. This yields 42 original segment files per participant, 420 per environment, and 1,260 in total. We do not extract extra overlapping training windows from these files. Selection rules and participant roles were saved before evaluating model outcomes.

Each environment assigns six participants to the source-training role, two to source calibration, and two to familiar testing, using a separate fixed hash ranking. For each of three folds, one complete environment is held out and the two remaining environments supply those source roles (Table 3). The target test includes all ten participants of the held-out environment. Source and target labels never enter fitting jointly, and no participant appears in more than one partition within a fold.

Table 3. File and participant counts in each leave-one-environment-out fold. Each file is one original activity segment.

Partition	Envs	People	Files
Training	2	12	504
Calibration	2	4	168
Familiar test	2	4	168
Held-out test	1	10	420

The source participants therefore change between training, calibration, and familiar tests. Target participants change as well. This is a test of transfer to a new room configuration and a new participant cohort. Since subjects 1–10, 11–20, and 21–30 belong to E1, E2, and E3 respectively, the dataset cannot separate a pure room effect from person-cohort differences. A matched-person, multi-room experiment would be required for that causal separation.

7. Transfer preprocessing and frozen evaluation

The MAT files contain packet-wise CSI and radio metadata. We port the author/Halperin scaling procedure for one transmit and three receive antennas [9, 10]. If H_p^{raw} is a packet's stored CSI, r_p the RSS-derived linear power and v_p its noise level in dB, the relevant scaling is

$$\begin{aligned} s_p &= \frac{r_p}{\frac{1}{30}\sum_{r,k} |H_{p,r,k}^{\mathrm{raw}}|^2}, \\ H_p^{\mathrm{sc}} &= H_p^{\mathrm{raw}} \sqrt{\frac{s_p}{10^{v_p/10} + 3 s_p}}. \end{aligned} \tag{12}$$

The RSS term is formed from nonzero chain RSSI values, with the source implementation's 44-dB offset and automatic-gain correction. A noise sentinel of −127 is replaced by −92 dB, following that implementation. These are receiver-processing conventions, not participant-specific body parameters or calibrated field measurements.

We use $20\log_{10}\max(|H_p^{sc}|, 10^{-12})$ and no phase features. Six bands of five adjacent exported subcarriers are averaged per receive antenna, giving 18 temporal streams. Linear interpolation onto 500 equally spaced packet-index positions gives a fixed-length representation. Thus duration and irregular timestamp gaps are not preserved; the horizontal coordinate is normalized sample position, not elapsed seconds. Six statistics and ten temporal-bin means per stream give a 288-feature representation for the two classical baselines.

Acquisition checked each selected file against the pinned repository's Git blob hash and recorded its SHA256. All 1,260 raw files and normalized representations were unique. Packet counts ranged from 827 to 1,438. The audit found no invalid zero-total-power or zero-RSSI packets, but 6,389 individual zero-amplitude coefficients required the declared logarithm floor. Three non-increasing timestamp steps were recorded; packet-index normalization does not reinterpret them as physical time. These checks assess input integrity rather than the completeness of the original acquisition protocol.

The two classical models use fixed settings: logistic regression with $C = 1$ and training-only standardization, and a 200-tree forest with minimum leaf size two. They are refitted independently in each fold using only its 504 training files. A single temperature per model/fold minimizes NLL on the 168 clean source-validation files. Calibration values and fitted-model comparisons are retained; target labels do not select hyperparameters, features, or temperatures.

Each test file is evaluated completely and under random or contiguous removal of exactly 50, 150, or 250 of 500 positions. Five hash-seeded masks are generated at each nonzero fraction, identically across its channels. Interpolation uses observed values only and holds exterior boundary values. Because band averaging and linear temporal interpolation commute, the shared-mask operation can be applied to the 18 band streams without accessing hidden samples. These are synthetic losses of normalized observations, not measured packet outages or a complete network failure model.

The experiment yields 31 conditions per domain, two test domains, three folds and two models: 372 metric rows and 109,368 per-file prediction records. These repeated views are not additional participants. Within each classifier, pooled target predictions use each of the 1,260 files once per mask seed and its fold-specific frozen model. Familiar tests contain 168 files per fold; their participants can recur as familiar tests in another fold, so pooled familiar evaluations are not independent replicates.

8. Cross-environment results

Table 4 reports complete-input accuracy before any imposed loss. Transfer accuracy varied across the three targets. Logistic regression scored higher on held-out E2 than on that fold's familiar test, whereas the other clean target comparisons were lower. Different cohort composition and finite samples prevent a universal ordering.

Table 4. Complete-input accuracy (%). E1–E3 identify the excluded target. LR/RF denote logistic regression/random forest.

Model / target	Familiar	Held out
LR / E1	47.0	34.8
LR / E2	36.9	39.0
LR / E3	42.9	36.2
RF / E1	46.4	43.6
RF / E2	45.8	37.1
RF / E3	44.6	36.2

Across held-out environments, complete-input accuracy was 462/1,260 (36.7%) for logistic regression and 491/1,260 (39.0%) for the forest. At 50% block loss, the five-mask mean accuracies were 25.5% and 28.8%. Random loss at the same fraction gave 31.8% and 36.6%. These averages weight the three equally sized target environments equally, and each participant contributes 42 files. They describe fixed lightweight baselines, not the best achievable performance on this dataset.

Table 5. Pooled held-out metrics. R50/B50 mean random/block loss at 50%. Accuracy is a percentage; F1 and NLL use native scales.

Condition	Acc.	Macro F1	NLL
LR clean	36.7	0.363	1.659
LR R50	31.8	0.330	1.780
LR B50	25.5	0.255	2.006
RF clean	39.0	0.392	1.443
RF R50	36.6	0.381	1.485
RF B50	28.8	0.293	1.851

The probability summaries in Table 5 pool target predictions within each seed, compute the metric, and then average the five corrupted-seed values. In particular, pooled macro F1 is not an average of per-environment F1 scores. Under block loss, NLL increased from 1.659 to 2.006 for logistic regression and from 1.443 to 1.851 for the forest. Multiclass Brier scores increased from 0.764 to 0.889 and from 0.713 to 0.835, respectively. Clean-source calibration therefore did not remove the probability errors associated with corrupted target inputs.

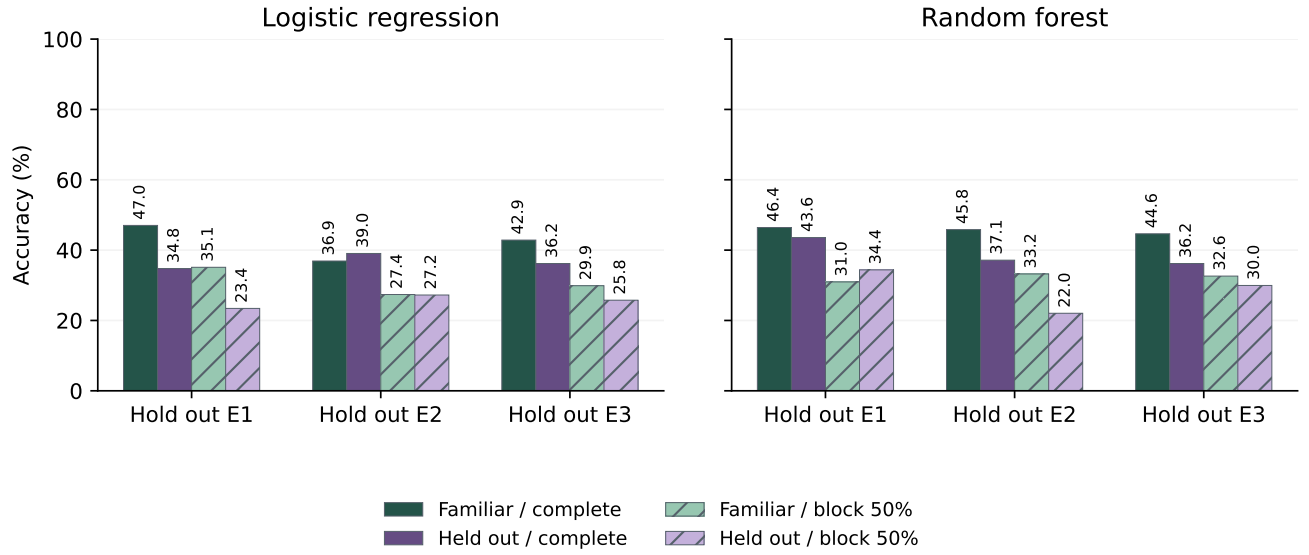


Figure 4. Classification accuracy on genuine recordings under the declared evaluation. Each group identifies the excluded target environment. Solid bars use complete inputs; hatched bars average five 50% block-loss masks. Familiar tests exclude training participants, while target tests exclude the whole environment. Counts are 168 familiar and 420 target files per fold. Bars are descriptive; they do not isolate a causal room effect.

Figure 5 shows that aggregate degradation was not uniform across activities. Logistic-regression walking recognition changed from 122/180 complete files to 123/900 masked views; the forest changed from 136/180 to 203/900. Counts in the corrupted panels are repeated views and must not be read as new independent trials. Sitting and lying were frequently confused even on complete inputs, while falls and pen-picking also generated errors. No activity-level safety claim follows from this evaluation.

9. Interaction contrast and conditional uncertainty

The sign of I_q in Equation 11 varies with target environment, model, missing fraction, and mask pattern (Figure 6). For 50% block loss, LR contrasts were −0.57, +2.29, and −2.55 percentage points for E1–E3. Forest contrasts were −6.29, +2.48, and −5.79 points. Hence the evaluated conditions do not support a uniform superadditive error increase. A negative contrast means the loss-associated error increment was smaller in the target test than in the familiar test; it does not imply robust target performance.

For conditional accuracy uncertainty, participant j contributes a mean correctness over its $n_j = 42$ files and S_q mask draws:

$$a_{j,q} = \frac{1}{n_j S_q} \sum_{i \in j} \sum_{s=1}^{S_q} \mathbb{1}\{\hat{y}_{i,q,s} = y_i\},$$
$$\widehat{A}_q = \frac{1}{30} \sum_{j=1}^{30} a_{j,q}. \tag{13}$$

Here $S_0 = 1$ for complete inputs and $S_q = 5$ for corrupted conditions. A stratified participant bootstrap samples ten IDs with replacement within each of the three target environments, retaining all files and masks of each sampled participant [14]. We use 10,000 draws with a fixed seed and 2.5th–97.5th percentiles (Table 6).

Table 6. Conditional participant-resampling ranges for held-out accuracy (%). They do not cover new environments or model retraining. R50/B50 follow Table 5.

Condition	Accuracy	95% range
LR clean	36.67	33.02–40.16
LR R50	31.78	28.30–35.22
LR B50	25.48	22.73–28.21
RF clean	38.97	35.00–42.94
RF R50	36.60	32.46–40.56
RF B50	28.79	25.37–32.00

The paired clean-to-block accuracy drops were 11.19 percentage points for logistic regression (conditional range 9.16–13.16) and 10.17 for the forest (7.32–12.79). For 50% random loss, the forest drop was 2.37 points with a range of −0.40 to 4.89. Thus this particular random-loss effect is less stable under the declared resampling than its block-loss counterpart.

These ranges preserve the within-person dependence of multiple trials and masks. They assume participant exchangeability within the sampled environments and condition on the already fitted models, selected subset, and mask draws. Training, split-selection, cohort-composition and

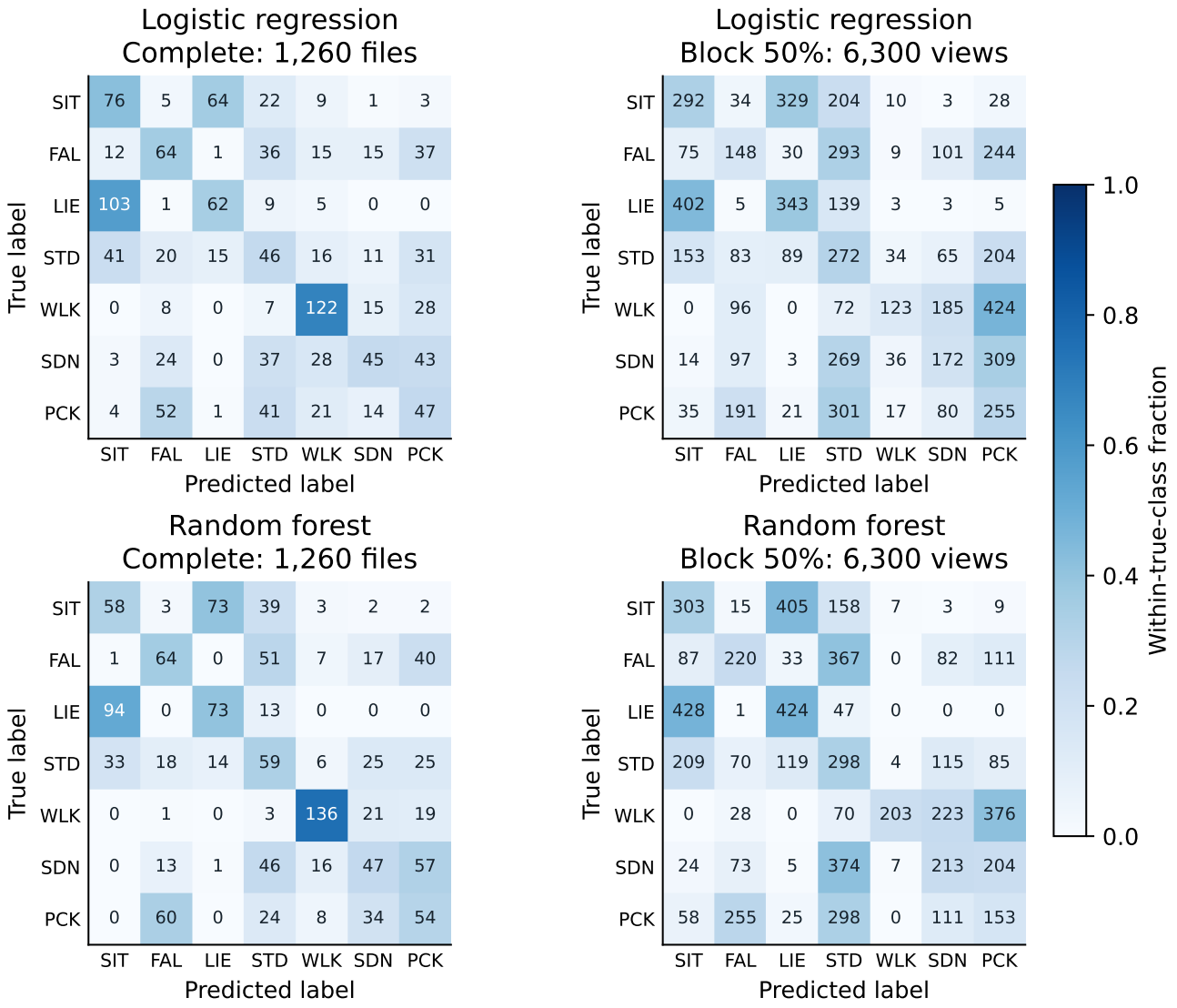


Figure 5. Held-out confusion matrices aggregated across the three fold-specific models of each classifier. Class keys follow Table 2. Complete panels contain 1,260 original-file predictions, with 180 files per class. Block panels pool five 50% masks, giving 6,300 views and 900 per class; these views are dependent repeats. Colors are within-true-class fractions and annotations are counts.

new-environment uncertainty are excluded. Three environments do not establish a population of deployment sites. We do not attach a significance test or a causal interpretation to the interaction heatmap.

10. Missingness-informed neural extension

The extension evaluates four fixed comparators on the same 1,260 files and participant roles: a compact CNN trained on complete source inputs, the same CNN with source-only loss augmentation, MIRA-CSI, and the official basic multivariate MiniRocket transform [15]. This extension was specified after the classical-baseline audit; it is an additional controlled comparison, rather than a prospectively registered study. It includes five training seeds (20261026–20261030) in each of the three folds, yielding 60 fitted runs. Target labels are used only for scoring.

MIRA-CSI denotes Missingness-Informed Recognition of Activities from Wi-Fi CSI. It adds explicit observation information and coverage-weighted pooling to a temporal CNN. Masks and temporal-gap representations have established precedents, including GRU-D [16]; the contribution assessed here is this particular CSI implementation and comparison, not the invention of missingness-aware learning. The amplitude-only clean and augmented CNNs compare training regimes on the same backbone. The initial MIRA-CSI comparison changes mask input, distance input and pooling together and therefore cannot isolate their individual benefits. Section 12 reports a subsequent component-removal experiment.

Let $B \in \mathbb{R}^{18\times500}$ denote the amplitude-band representation. Its channel means and standard deviations are estimated only from source-training



Figure 6. Descriptive difference-in-differences error contrast $100I_q$ in percentage points. Warm colors indicate a larger loss-associated error increase in the held-out domain; cool colors indicate a smaller increase. Zero is the additive reference on this error-rate scale. Each value uses the same clean and masked test files within its domain and averages five masks. Room/cohort confounding and only three environments limit interpretation.

files. For repaired standardized amplitudes $\hat{B}$, the additional gap channel and network input are

$$d_t = \frac{\min_{s:m_s=1} |t-s|}{500}, \qquad Z_t = [\hat{B}_{:,t}; m_t; d_t]. \quad (14)$$

Distance is measured in normalized packet positions; it is not elapsed time. It is zero at observed positions by construction and can use an observation on either side of a gap. On this integer grid, $m_t = \mathbb{I}[d_t = 0]$, so the distance channel already determines the mask exactly. Removing the explicit mask channel therefore tests an encoding/parameterization choice rather than removing observation-status information. This offline support-distance feature differs from the elapsed-since-previous-observation representation in GRU-D and the retained-sample age relative to a session decision in Zakeri et al. [16, 20]; it is not a causal streaming feature. Three convolutions use 32, 64 and 128 channels, kernel lengths 7, 5 and 3, stride two, GroupNorm with eight groups, and GELU. If h_j is an encoded temporal feature and $\bar{m}_j$ the adaptively averaged observation mask at that resolution, pooling is

$$w_j = \max(\bar{m}_j, 0.05), \qquad g = \frac{\sum_j w_j h_j}{\sum_j w_j}. \quad (15)$$

The weights describe mask-bin coverage, not calibrated confidence or a physical RF reliability estimate. GroupNorm aggregates statistics across time, so the weights are a coverage heuristic rather than the exact observed fraction of each feature's full dependency set. Dropout (0.2) and a seven-logit linear head complete the network. MIRA-CSI has 40,871 trainable parameters, compared with 40,423 for the amplitude-only CNN. The ordinary CNN uses uniform temporal averaging.

All CNNs use AdamW (learning rate 10^{-3}, weight decay 10^{-4}), batches of 64, at most 60 epochs, and gradient clipping at norm five. Checkpoints minimize clean source-validation cross-entropy, with patience 12 and minimum improvement 10^{-4}. In both augmented variants each training view remains clean with probability 0.5; otherwise loss pattern, fraction (0.1/0.3/0.5) and repair method are drawn uniformly from their predefined choices. The paired variants use the same independent augmentation-generator draw order for each seed and fold. No target-domain selection is performed. The package retains validation histories and every trained state.

MiniRocket uses the pinned official Numba implementation with 10,000 requested features (9,996 produced), training-only feature scaling without centering, and ridge classification with $\alpha = 1$. This compiled comparison is distinct from the earlier 84-feature compatibility check. For every neural or ridge model, a single temperature fitted to clean source-validation logits is frozen on all test conditions.

Three offline repair operators preserve observed values: linear interpolation with held boundaries, nearest observed value (earlier observation on ties), and source-training band mean. These operators do not use hidden values or target labels; interpolation may use later observations and therefore does not establish causal streaming operation. The extended

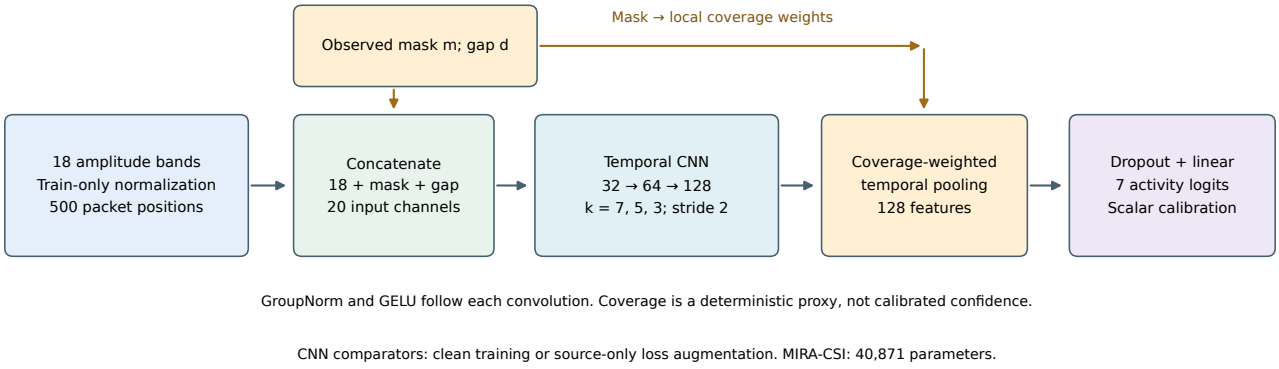


Figure 7. Architecture of the implemented MIRA-CSI classifier. Observation mask and distance supplement 18 amplitude bands; mask coverage also weights temporal pooling. The amplitude-only CNN comparators use the same convolutional backbone. These comparators do not isolate the three added components individually. This is an algorithm diagram, not a body reconstruction or a measured RF field.

comparison focuses on 50% random and contiguous loss. A documented compute amendment pairs each training seed with the same mask seed for the primary cross-model comparison, retaining all five fitted seeds and all three folds. The CNNs additionally retain the original five-mask grid as supplementary sensitivity data. Previously completed full-grid MiniRocket output is also retained.

Primary summaries pool the three held-environment prediction sets separately for each seed, then report the mean and sample standard deviation across the five paired training/mask seeds. Familiar summaries pool fold-specific source-test predictions; some familiar files recur across folds and are not independent observations. These standard deviations combine training and mask variation, conditional on the fixed recordings and partitions; they are not population confidence intervals. Repair error is evaluated only at synthetically hidden values using root-mean-square error on the scaled dB amplitude representation.

11. Extended model and repair results

The independently checked extension retains all 60 fitted runs, complete probability vectors, validation histories and checkpoints. Table 7 and Figure 8 report the matched primary conditions. Each seed pools three excluded-environment test sets; the five seed summaries describe variation conditional on this selection, rather than five independent datasets. The compiled MiniRocket reference achieved 53.1% clean held-environment accuracy and 19.6% under 50% block loss with linear repair. MIRA-CSI achieved 23.7% and 23.0%, respectively. The CNN comparators and all repair choices are retained; the extension does not impose a target accuracy or establish universal superiority of a model family.

Relative to the augmented amplitude-only CNN, adding the MIRA inputs and pooling changed pooled held block-loss macro F1 by +0.018 on average (SD 0.023); the paired difference was positive in 5 of five seeds. This is a comparison of the combined modification. It does not isolate mask, distance and pooling effects individually, and it is not a significance test.

Table 8 separates signal reconstruction from downstream recognition and probability quality. A lower hidden-value RMSE is not equivalent to a lower classification error: smoothing and class-discriminative variation can respond differently. The saved NLL, Brier and ECE results allow probability quality to be examined without treating a plausible activity label as a calibrated guarantee.

The earlier 84-feature NumPy compatibility check remains in the reproduction archive as a bounded execution audit. It is superseded here by an official compiled transform with 9,996 features and five fitted seeds; its scores and runtime are not pooled with the extension. All new scores are secondary analyses of recorded activities, not demonstrations of live pose reconstruction.

11.1. Conditional participant-level sensitivity

A post-hoc analysis of frozen predictions separates the primary five paired-seed summaries from a full mask-grid sensitivity analysis. For each of the three neural configurations, all five training seeds are crossed with all five loss-mask seeds, yielding 25 cells per corrupted condition and five cells for complete inputs. No weights, partitions or temperatures are refitted. Within each cell, predictions from the three held environments are pooled before scoring; the reported estimand averages those cell-level scores, rather than first averaging probability vectors.

Table 7. Extended held-environment accuracy (%). Means ± sample SD over five paired training/mask seeds; each seed pools 1,260 excluded-environment predictions. R50/B50 denote random/contiguous 50% loss with linear repair. MR is the compiled 9,996-feature MiniRocket reference.

Model	Complete	R50	B50
CNN	24.5 ± 2.2	24.2 ± 2.0	22.2 ± 2.1
CNN + aug.	22.5 ± 1.8	22.2 ± 1.6	20.8 ± 1.3
MIRA-CSI	23.7 ± 2.3	22.7 ± 2.2	23.0 ± 1.3
MR (9,996)	53.1 ± 0.3	49.5 ± 0.6	19.6 ± 0.9

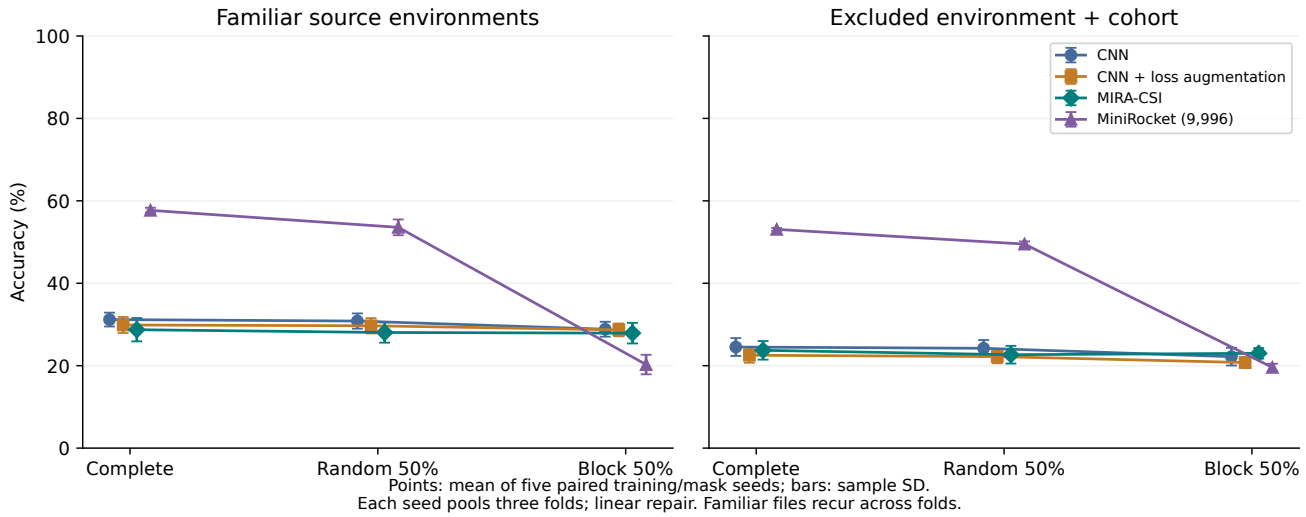


Figure 8. Measured five-seed model comparison. Points are means and bars are sample standard deviations across paired training/mask seeds, with linear repair for corrupted inputs. Held results pool 1,260 distinct files per seed; familiar results pool fold-specific predictions and reuse some files across folds. These bars do not represent uncertainty over new buildings or populations.

Table 8. Repair and downstream performance at 50% loss. RMSE is pooled over synthetically hidden amplitude values in scaled dB units. F1 and NLL average five paired seeds after pooling three target folds. MIRA and MR denote MIRA-CSI and MiniRocket (9,996). Repair ground truth is used only for scoring.

Loss	Repair	RMSE	MIRA F1	MR F1	MIRA NLL	MR NLL
Random	Linear	1.03	0.209	0.487	2.140	1.593
Random	Nearest	1.19	0.209	0.487	2.137	1.586
Random	Source mean	3.43	0.198	0.150	2.059	3.353
Block	Linear	2.33	0.217	0.156	2.156	5.613
Block	Nearest	2.58	0.216	0.180	2.137	4.517
Block	Source mean	3.42	0.182	0.129	2.541	5.015

For 10,000 bootstrap replicates, ten participant clusters are sampled with replacement independently within each of the three fixed target environments. Each cluster retains its 42 recordings and every saved seed/mask view. Identical resampling weights are applied to all compared models and conditions. The 2.5th and 97.5th percentiles of paired metric differences give the conditional ranges in Table 9. These ranges condition on the observed rooms, recordings, fitted models, calibration and mask grid; they do not quantify uncertainty over new environments or model retraining. This is an exploratory sensitivity analysis, not a prospectively specified confirmatory test, and its intervals are not adjusted for multiple comparisons.

Under contiguous loss, the full-grid macro-F1 difference is smaller than the primary five-pair difference (0.0136 versus 0.0182). The positive conditional classification range is small and unadjusted; the NLL range includes zero. Complete-input and random-loss F1 ranges also include zero. Consequently, this analysis supports at most a narrow,

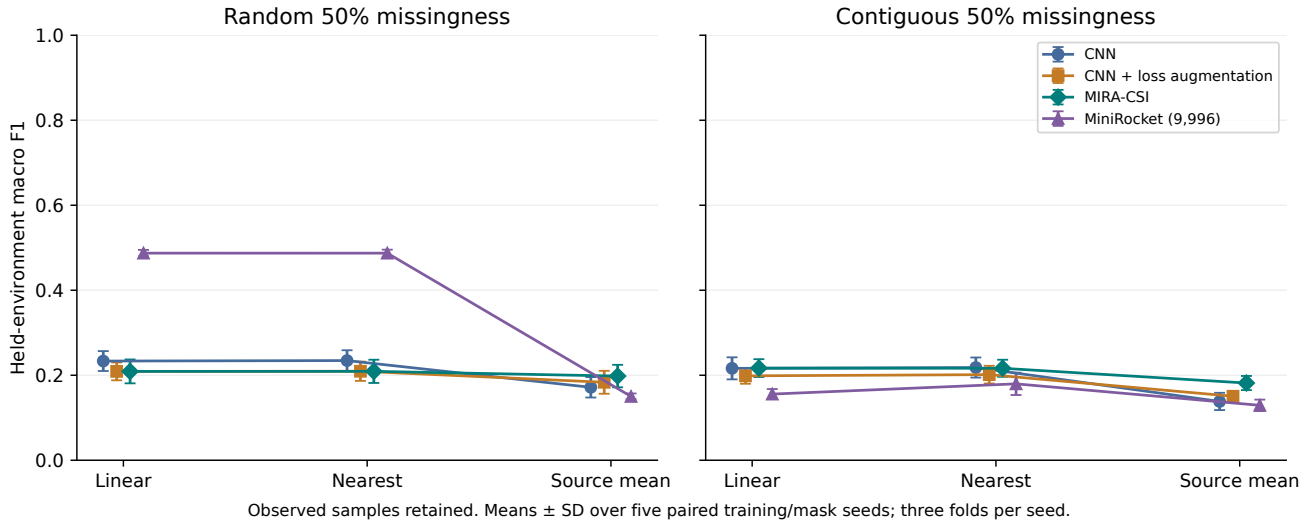


Figure 9. Downstream recognition under three repairs of exactly matched missing positions. Macro F1 pools the three held-environment sets for each seed. Error bars are SD across five paired training/mask seeds. Models and clean-source calibration remain frozen across repair choices; later observations may be used by these offline operators.

Table 9. Full-grid paired differences: MIRA-CSI minus augmented CNN. Each cell shows the point estimate and a conditional 95% participant-bootstrap percentile range. Accuracy differences are percentage points; macro F1 and NLL differences use their original scales. Positive accuracy/F1 differences and negative NLL differences favour MIRA-CSI. R50/B50 use linear repair.

Input	Δ Accuracy (pp)	Δ Macro F1	Δ NLL
Complete	+1.19 [−0.14, 2.57]	+0.0107 [−0.0043, 0.0265]	−0.0354 [−0.0892, 0.0174]
R50	+0.44 [−1.00, 1.88]	+0.0003 [−0.0158, 0.0164]	+0.0149 [−0.0358, 0.0688]
B50	+1.84 [0.68, 3.01]	+0.0136 [0.0005, 0.0275]	−0.0410 [−0.0985, 0.0158]

dataset-conditional classification difference under the declared block-loss procedure. It establishes neither general probability-quality improvement nor the separate value of any MIRA-CSI component. Per-environment scores, seed-grid summaries, all declared contrasts, resampling weights and probability-file hashes are retained with this revision.

12. Component controls and representation sensitivity

12.1. Frozen component-removal experiment

A follow-up experiment isolates the explicit mask-input channel, nearest-support-distance input, and coverage-weighted pooling of MIRA-CSI. Four controlled configurations retain the same 20-channel architecture, 40,871 nominal trainable parameters, initialization at each seed, source partitions, augmentation draws, optimizer, checkpoint rule and clean-source temperature calibration. The reference retains all three components. The other configurations respectively zero the mask-input channel, zero the distance-input channel, or replace coverage weighting with uniform temporal averaging. Zeroing an input retains its nominal parameters but removes its data-dependent gradient. In particular, removing the explicit mask input does not remove all missingness information: distance and coverage weighting remain available.

Two standard residual 1D CNNs provide separate architecture-and-capacity comparators, with and without source loss augmentation. Each has three residual blocks of 64/128/128 channels, convolutions of lengths 7/5/3, GroupNorm and GELU, followed by eight-bin temporal averaging and a seven-class head; each has 511,111 parameters. Their architecture is a reference design rather than a new MIRA component. Architecture and parameter count change together, so these models are not component-level ablations and do not isolate a capacity effect. The residual models use the same source-only optimizer, epoch-selection and calibration settings as the compact networks. CUDA-safe explicit temporal-bin means implement the declared pooling operation.

The six configurations, three outer folds and five training seeds produce 90 source-fitted models. All checkpoints and calibration parameters were frozen and hashed before this follow-up evaluation. Each model is tested on complete input and the full five-mask grid of random/contiguous 50% loss under all three repairs: 31 conditions on both

Table 10. Frozen follow-up held accuracy (%). Corrupted conditions average all five mask realizations within each training seed; cells show mean ± sample SD across five training seeds after pooling the three held sets. R50/B50 use linear repair. All four MIRA configurations have 40,871 nominal trainable parameters; both residual controls have 511,111.

Configuration	Complete	R50	B50
MIRA full reference	23.73 ± 2.26	22.68 ± 2.03	22.56 ± 1.28
Zero explicit mask input	23.41 ± 1.03	23.22 ± 1.23	21.90 ± 0.53
Zero distance input	24.03 ± 1.87	22.85 ± 1.87	22.77 ± 1.02
Uniform temporal pooling	24.17 ± 2.54	23.22 ± 1.86	21.30 ± 1.57
Residual CNN, clean training	33.83 ± 0.82	33.33 ± 0.87	26.72 ± 0.67
Residual CNN, augmented	30.25 ± 1.65	30.09 ± 1.70	25.72 ± 1.35

familiar and held sets. This retains 5,580 metric rows and 1,640,520 probability vectors. These are repeated views of the same recordings, not additional participants. Independent arithmetic reconstruction reproduced every metric row, with maximum absolute difference 4.44×10^{-16}. A separate CPU audit loaded every frozen checkpoint and reproduced 15,120 source-validation prediction views without a classification-label mismatch; the maximum CPU/T4 logit difference was 6.20×10^{-6}.

The follow-up is exploratory: its design responds to results already observed on this dataset. Freezing the new fits prevents target-dependent selection within this execution, but does not make the historical target data an untouched external confirmation set. For a corrupted condition, held predictions are first pooled across the three folds for each training/mask-seed cell. Scores are then averaged across five masks within each training seed, and the mean and sample SD are reported across the five training seeds. This differs from the initial paired-seed summary in Table 7.

The larger residual clean-training control reaches 33.83% held accuracy on complete input and 26.72% under block loss (Table 10, Figure 10); its augmented counterpart reaches 30.25% and 25.72%. The residual architectures improve these particular compact-network results, but changing architecture and capacity together does not identify the cause of that difference or establish a generally optimal model. The full MIRA reference reproduces its original clean accuracy. Its block value here averages the full mask grid and therefore differs from the earlier five paired-mask summary. SD across these five mask-averaged seed summaries is not the same quantity as SD across the original five diagonal training/mask pairs. For conditionally independent, identically distributed mask draws within a fixed fit, averaging five masks reduces that fit's mask-variance component by a factor of five; this is not a rule for the total variability across trained models.

For component contrasts, 10,000 paired participant-cluster bootstrap draws sample ten participants with replacement within each fixed environment and retain their complete seed/mask grid. The full model minus each of the three removal controls under linear B50 defines one three-comparison macro-F1 family. Table 11 reports ordinary 95% percentile ranges and Bonferroni-adjusted 98.333% marginal percentile ranges for that family. These approximate conditional sensitivity ranges remain exploratory, depend on within-environment participant exchangeability and fixed fitted models, and do not cover new rooms, retraining or all secondary outcomes. The resampling unit is the participant rather than an individual mask view.

The explicit mask-input contrast is inconclusive in this condition. Omitting the distance input gives a small higher mean macro F1 than the full reference, including under the declared conditional three-contrast adjustment; however, the paired difference changes direction across training seeds (Table 11). It is not a stable advantage across every initialization. Coverage weighting has a positive unadjusted range, but its adjusted range includes zero (Figure 11). Thus the experiment does not support presenting all three MIRA components as individually beneficial. These condition-specific findings neither prove that a component has no value elsewhere nor validate the full architecture as superior to the reference transform. Different complete-input scores after channel removal are also compatible with retraining under a different parameterization: even a constant clean mask channel can change convolutional activations and the optimization trajectory, including padding-boundary effects. No physical mechanism is identified by that score difference alone.

12.2. Source-selected representation of the reference transform

A separate clean-input sensitivity study strengthens the known MiniRocket comparator without attributing that algorithm to MIRA-CSI. Before fitting, five candidate input representations were fixed: the original bands, one scalar z-score per

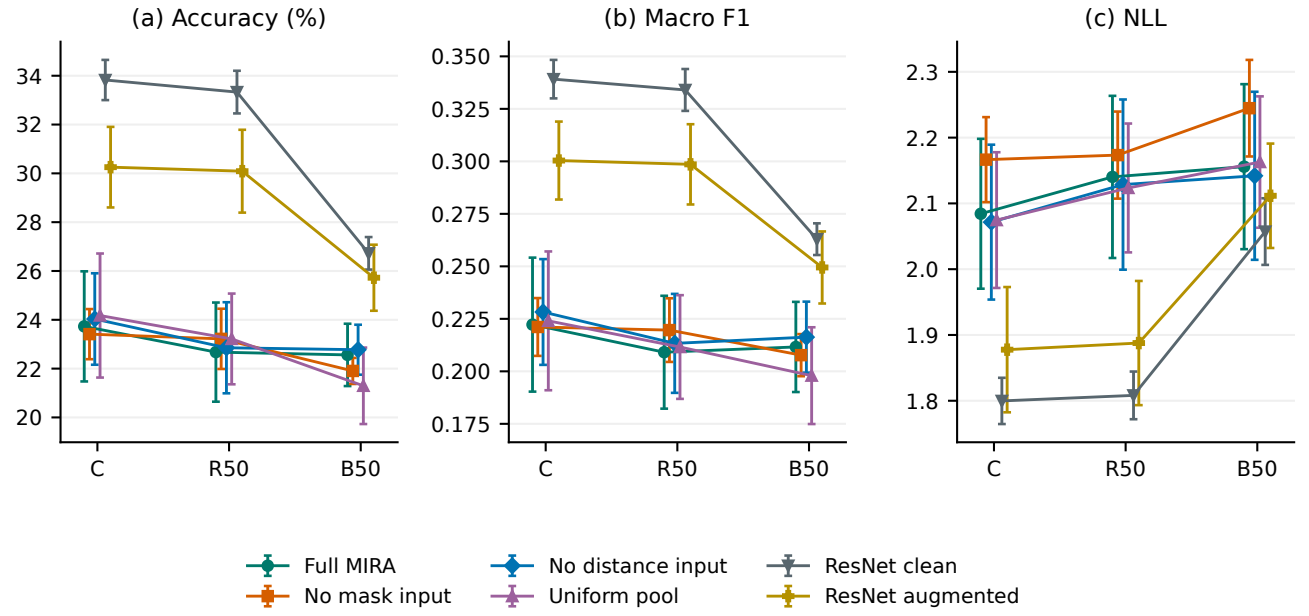


Figure 10. Frozen follow-up comparison of accuracy, macro F1 and source-temperature-calibrated NLL. C is complete input; R50/B50 are random/contiguous 50% loss with linear repair. Metrics pool 1,260 held files within each training/mask cell, then average masks within training seed. Points show means and bars sample SD across five training seeds on fixed recordings and partitions; they are not confidence intervals for new environments. Higher accuracy/F1 and lower NLL are preferred.

Table 11. Component contrasts under contiguous 50% loss with linear repair. Differences are full MIRA minus the named control in pooled macro F1. The second column adds sample SD of paired, mask-averaged differences across five training seeds; the final column counts positive seed differences. Percentile ranges share identical participant draws, condition on the fixed fitted models, and address only the three contrasts shown.

Control	Δ F1 $\pm$ SD	Conditional 95%	Adjusted 98.333%	Positive
Zero explicit mask input	$+0.0039 \pm 0.0126$	$[-0.0074, 0.0155]$	$[-0.0096, 0.0180]$	3/5
Zero distance input	-0.0047 ± 0.0117	$[-0.0080, -0.0015]$	$[-0.0086, -0.0007]$	2/5
Uniform temporal pooling	$+0.0137 \pm 0.0190$	$[0.0001, 0.0275]$	$[-0.0023, 0.0302]$	4/5

recording, a temporal z-score per band, source-training per-band 0.1/99.9-percentile winsorization, and log-mean-power bands. The last candidate averages scaled squared amplitudes before applying $10 \log_{10}$ and retains the same 500-position resampling. It uses relative scaled power, not calibrated dBm. Source-only inspection motivated this numerical sensitivity check; changing the transform is not evidence of an architectural contribution.

Within each outer fold, representation selection uses only that fold's source-validation macro F1 at seed 20261026. Validation scores are never pooled across outer folds for selection, because another fold can contain the current target environment. The selected representation is then held fixed across all five seeds. E1 selects per-band temporal standardization; E2 and E3 select scalar recording standardization. MiniRocket remains the official 9,996-feature multivariate transform, with the same source-only feature scaling, ridge coefficient and temperature-fitting rule. All 30 reference/selected fits were sealed before any new held scoring; six seed-one fits were reused and 24 additional fits were trained.

The pooled accuracy difference is 4.37 percentage points (Table 12). Per-environment mean accuracy changes from 57.24% to 66.43% in E1, 50.67% to 50.43% in E2, and 51.33% to 55.48% in E3. The gain is therefore not shared by every environment. Independent checks reconstructed scores from 12,600 saved held probability vectors and verified all 30 checkpoint hashes without repeating target inference. Recording standardization in this stage uses complete recordings; this clean-only experiment establishes neither missing-input robustness nor an information-safe standardization procedure under hidden-value loss. No inference about a new MIRA mechanism follows from the MiniRocket improvement.

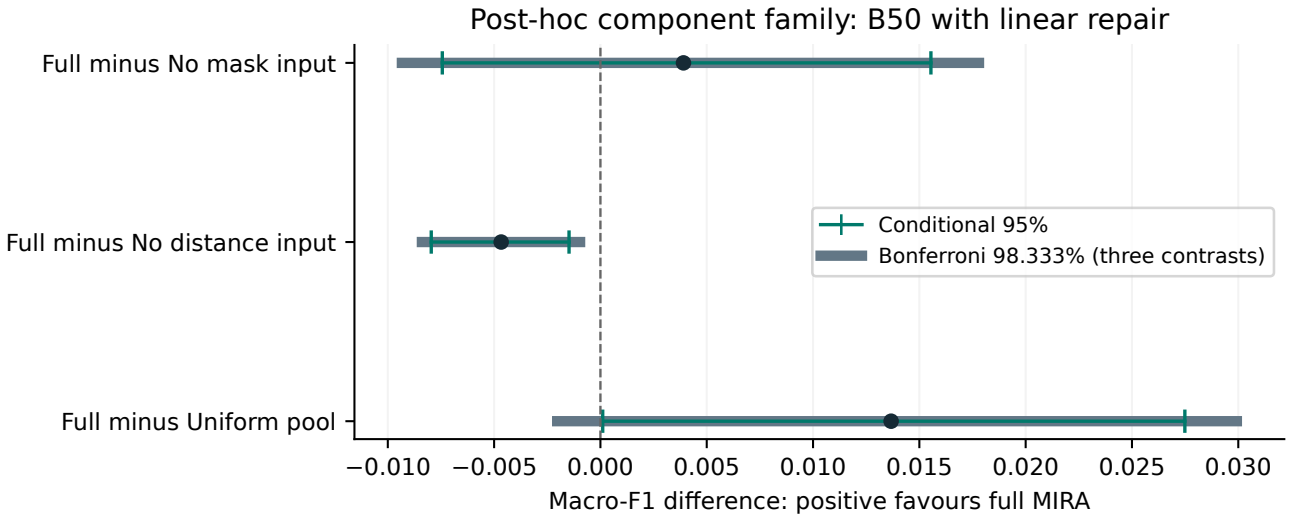


Figure 11. Paired component differences under linear B50. Thin green segments with terminal ticks mark unadjusted 95% participant-bootstrap percentile ranges; thick gray segments mark 98.333% ranges adjusted for the three displayed contrasts. Each point is full MIRA minus a removal control, so positive values favour the full model. All ranges use the same 10,000 participant draws within the three fixed environment/cohort strata. This exploratory post-hoc analysis does not measure uncertainty over new environments or model retraining.

Table 12. Clean-input MiniRocket representation sensitivity. Means ± sample SD across five seeds; metrics pool 1,260 held files per seed. The selected representation is chosen independently from each outer fold's source validation. This is a reference-transform improvement, not a new MIRA model.

Input representation	Accuracy (%)	Macro F1	NLL	Brier
Original	53.08 ± 0.33	0.531 ± 0.003	1.395 ± 0.049	0.639 ± 0.008
Source-selected	57.44 ± 1.04	0.574 ± 0.009	1.259 ± 0.040	0.586 ± 0.013

The selected pipeline's seed SD is 1.04 percentage points, compared with 0.33 for the original representation. The representation choice is fixed from one source-validation seed; its stability across alternative selection seeds has not been measured. Furthermore, the CNNs were not subjected to the same five-candidate input-selection procedure. This sensitivity result therefore compares two declared MiniRocket pipelines and does not isolate an architectural cause for their differences from the neural models.

13. Probability quality and selective ranking

A descriptive post-hoc audit examines the already frozen neural and ridge predictions without fitting another model, temperature or threshold. Table 13 and Figure 12 retain the initial 60-fit package with five paired training/mask seeds; they are not the full-grid follow-up. Figure 10 and the follow-up score files report the 90-fit comparison, including both residual controls, using mask-averaged seed summaries. Table 13 reports proper scores alongside macro F1 and ECE. Because every pooled held set is balanced over seven classes, the uninformative reference distribution $p_k = 1/7$ has

$$\mathrm{NLL}_{\mathrm{unif}} = \log 7 \approx 1.946, \qquad \mathrm{Brier}_{\mathrm{unif}} = 1 - \frac{1}{7} \approx 0.857. \tag{16}$$

This is a proper-score comparison, not a claim that uniform probabilities identify activities. Tied uniform confidence also cannot define a meaningful rejection ranking without an additional rule.

The mean NLL and Brier values for all three compact neural configurations exceed this uninformative reference on both complete and block-masked target inputs. Their accuracy above the balanced-class chance level therefore does not establish useful probabilistic confidence. MiniRocket improves both proper scores on complete inputs, but its block-loss NLL and Brier are markedly worse. Clean-source temperature scaling does not transfer a calibration guarantee to the excluded environment and cohort.

At 80% retained coverage under block loss, mean selective error was 76.8% for MIRA-CSI and 79.8% for MiniRocket. Thus, retaining high-confidence predictions is not sufficient to guarantee low error in these target conditions. Figure 12 records the complete coverage range rather than choosing a

Table 13. Held-environment probability quality and macro F1. C: complete input; B50: contiguous 50% loss with linear repair. Values are means ± sample SD over five paired seeds after pooling three folds. NLL, summed multiclass Brier and ECE are lower-is-better; F1 is higher-is-better. The uniform distribution is an untrained proper-score reference, not a useful activity discriminator.

Model	Input	NLL	Brier	ECE	Macro F1
CNN	C	2.091 ± 0.053	0.915 ± 0.020	0.197 ± 0.026	0.235 ± 0.025
CNN	B50	2.183 ± 0.046	0.941 ± 0.017	0.227 ± 0.026	0.216 ± 0.026
CNN + aug.	C	2.120 ± 0.059	0.928 ± 0.022	0.205 ± 0.032	0.212 ± 0.023
CNN + aug.	B50	2.199 ± 0.052	0.951 ± 0.018	0.230 ± 0.024	0.199 ± 0.019
MIRA-CSI	C	2.084 ± 0.114	0.908 ± 0.030	0.184 ± 0.041	0.222 ± 0.032
MIRA-CSI	B50	2.156 ± 0.130	0.927 ± 0.031	0.207 ± 0.042	0.217 ± 0.021
MR (9,996)	C	1.395 ± 0.049	0.639 ± 0.008	0.124 ± 0.009	0.531 ± 0.003
MR (9,996)	B50	5.613 ± 0.430	1.370 ± 0.037	0.633 ± 0.028	0.156 ± 0.012
Uniform reference	–	1.946	0.857	–	–

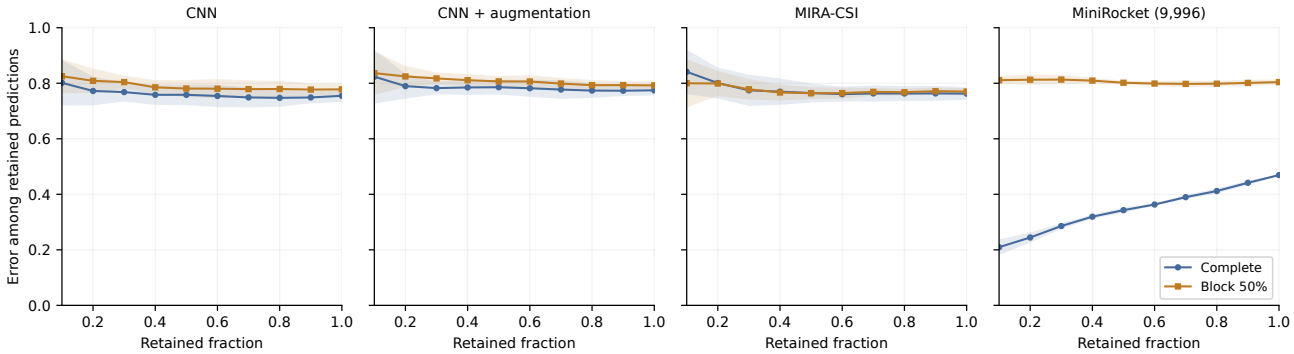


Figure 12. Descriptive risk–coverage curves of frozen models on held environments. At each retained fraction, predictions are ranked by maximum calibrated probability within that test condition. Lines average five paired seeds; shading is sample SD, not a population confidence interval. Confidence ranking selects observations; labels are used only to calculate retained error. The curves do not validate a deployable threshold.

favorable operating point. These are conditional error descriptions, not a validated selective-risk bound.

14. Discussion

The completed comparisons show why model capacity, missingness handling and transfer must be assessed separately. The wider MiniRocket has stronger clean recognition, but its long-gap probability errors are larger in the original-representation loss experiment. Source-selected recording normalization improves its clean accuracy, without testing its loss robustness. Residual capacity improves on the compact CNNs in the follow-up, yet the examined models remain weak on held environments. These results cannot support a claim that high-capacity deep learning universally fails after achieving high source accuracy. All transfer statements use within-dataset partitions.

Structured loss changes information before the classifier sees it. A long interpolated gap is determined by its observed boundary samples; it cannot uniquely recover an unobserved transition. The feature set includes temporal-bin means and first-difference statistics, both affected by that operation. This gives a plausible processing explanation for degradation, but a causal account of the individual class failures requires feature-specific ablations and further controlled acquisition. The extension compares three fixed repair operators but does not isolate every representation effect. These experiments do not establish that all CSI representations respond in the same way.

The interaction results also refine the original hypothesis. Additional error under loss is not consistently larger in the target domain. Low clean accuracy leaves different scope for further error, and cohort and class-specific confusions can alter the contrast. Reporting both signs is necessary; selecting only the positive cells would overstate the evidence. Future comparisons should include additional frozen baselines and independently collected environments before generalizing this pattern.

Target-data access defines a separate comparison axis. Standard domain-adversarial training uses labelled source observations and unlabelled target observations [17]. Such a method would require an explicitly separate adaptation protocol here, because the present fitting and calibration exclude target observations. An adversarial alignment restricted to the source environments could preserve that access constraint, but it has not been evaluated in this study. Neither access to target data nor an adaptation method constitutes a guaranteed performance upper bound.

The component-removal follow-up constrains the model contribution more directly than a bundled CNN/MIRA comparison. Under the declared block-loss procedure, the explicit mask-input contrast is inconclusive, the distance input has a small adverse classification association relative to its removal control, and the coverage-pooling contrast does not exclude zero after the three-comparison adjustment. These are useful design findings rather than evidence that every proposed component must help. The analysis is conditional on this training procedure and dataset; it neither proves universal component failure nor isolates a physical cause. Learned CSI recovery methods offer additional comparisons [18, 19], but their training access, reconstruction objectives and tasks must be matched before importing performance claims.

The augmented residual control also scores below its clean-trained counterpart in the reported complete, random-loss and block-loss conditions. This observation does not identify a cause. The training distribution mixes clean views with several loss fractions and repairs, whereas checkpoint selection minimizes clean source-validation NLL. In the declared sampling scheme, contiguous 50% loss has probability 1/12 per training view, or 1/36 with a particular repair. A mismatch between that mixture, the selection objective and the evaluated condition is a plausible hypothesis, not an experimentally isolated explanation. Mean repair can also introduce constant interiors and boundary changes, but the present comparison does not attribute the performance difference to those features. Testing these hypotheses would require source-selected augmentation and checkpoint-objective controls with a separate frozen evaluation.

Calibration and rejection require their own validation. NLL and Brier scores worsened under block loss despite a temperature fitted only to clean source validation. Saved selective-error metrics use 134/168 accepted familiar files (79.76%) and 336/420 target files (80%), ranked separately within condition. Such ranking describes a risk–coverage relationship; it does not produce a threshold guaranteed to control error on a new site. Deployment would require a separate validation protocol and a frozen operating rule.

The scope remains activity classification. Conceptual body meshes, colored wavefronts, and water-content diagrams explain concepts; none is a measured pose, temperature map, RF solution or physiological estimate. Inputs are presegmented, single-activity recordings with known boundaries. Continuous-stream detection and unseen activities are not evaluated. Different cohorts per environment also prevent separating room and person effects. It does not validate unrestricted sensing in homes, universities, companies, or multi-person scenes.

15. Reproducibility and conclusion

The source revision, file hashes, participant roles, trial IDs, mask seeds, model settings, temperatures, probability vectors, score tables, and bootstrap indices are retained. Independent calculations reproduced all 372 transfer metric rows from 109,368 predictions; all 1,260 input SHA256 hashes were rechecked. Radio scaling was independently spot-checked on nine files. These checks do not validate unrecorded acquisition conditions or original labels.

The package contains executable analyses and numerical outputs; recordings remain linked to the public sources [8, 9]. The earlier NTU-Fi loss-only pilot appears in Appendix A, uses a different dataset and label set, and is not pooled with the main study. The MIRA-CSI software, model card and citation metadata are publicly available at the author repository [21]. This secondary analysis involved no new participant recruitment or sensing hardware.

The initial extension and follow-up retain separate frozen protocols and outputs. The latter contributes 90 checkpoint-verified fits, 5,580 independently reconstructed metric rows and 1,640,520 saved probability vectors; its source-only archive is preserved separately from target evaluation. The representation sensitivity contributes 30 sealed fit configurations and 12,600 independently checked clean held predictions, with explicit reuse of previously fitted seed-one models. These counts describe computational artifacts and repeated evaluations, not an increase in the 30-participant sample. Script versions, input and prediction hashes, exact masks and participant-bootstrap draws accompany the corresponding result packages.

Across three held-out environments and 30 participants, complete-input accuracy was 36.7% for logistic regression and 39.0% for random forest; 50% block loss reduced these to 25.5% and 28.8%. Class confusions and probability errors revealed additional limitations. Interaction contrasts had both signs, and participant-grouped resampling quantified conditional sensitivity. MIRA-CSI is an executable,

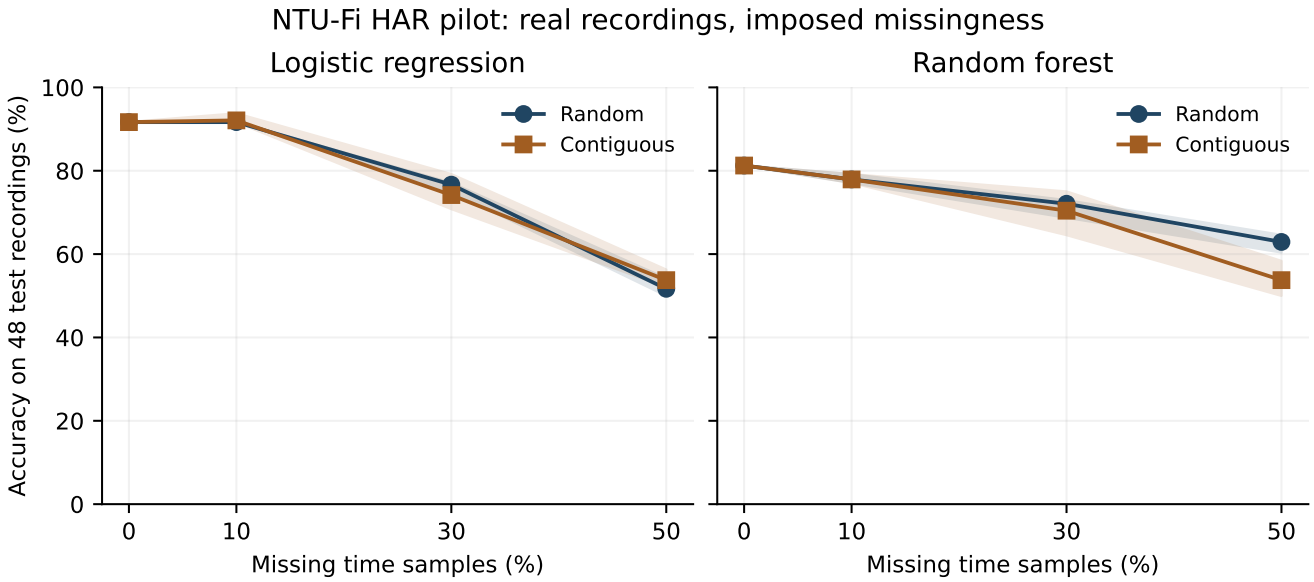


Figure 13. Computed pilot accuracy on the same 48 test recordings after linear interpolation. Lines average five mask realizations; shading spans their minimum and maximum. This band measures mask variability, not a confidence interval. These curves derive from genuine recordings and are separate from the explanatory 3D illustrations.

documented research model whose component controls reveal limited and condition-specific evidence, including a small improvement when distance is omitted. The larger clean-trained residual control reaches 33.8% held accuracy, while source-selected preprocessing improves the known MiniRocket comparator to 57.4% on complete input. These results support reporting component effects, representation choices and probability quality together. Independent replication with matched participants across more rooms and deployment-relevant acquisition remains necessary before broader performance claims.

A. Separate NTU-Fi pipeline pilot

A prespecified subset of NTU-Fi HAR exercises the loss-only evaluation. This pilot uses the released train/test partition and does not evaluate held-out environments.

The SenseFi public archive [5] contains 936 training and 264 test files, with six activity classes. A fixed SHA256 ranking selected 12 training, four validation, and eight test files per class: 72/24/48 files in total. Validation came exclusively from the released training partition. File checks found no non-finite values or exact duplicates in the selected inputs. Subject, session, and room identifiers were unavailable in this processed release.

Each file provides a 342×2000 amplitude array. Following the released loader, every fourth time sample was retained. Three antenna streams were summarized over six subcarrier bands using six statistics and ten temporal-bin means, yielding 288 features. Logistic regression ($C = 1$; training-only standardization) and a 200-tree random forest (minimum leaf size two) were fitted once. Temperature scaling used only the 24 clean validation files.

At each loss fraction, exactly 50, 150, or 250 of 500 time positions were masked across all channels. Uniform random masks and one contiguous block were generated with five fixed seeds. Linear interpolation used observed values only, with held boundary values. Test labels did not influence model fitting, calibration, or mask generation.

Table 14. Pilot accuracy (%). LR: logistic regression; RF: random forest. Corrupted conditions average five masks on the same 48 files.

Input condition	LR	RF
Complete	91.7	81.3
Random 10%	91.7	77.9
Random 30%	76.7	72.1
Random 50%	51.7	62.9
Block 10%	92.1	77.9
Block 30%	74.2	70.4
Block 50%	53.8	53.8

Complete-input accuracy was 44/48 (91.7%) for logistic regression and 39/48 (81.3%) for the forest. At 50% contiguous loss, both averaged 53.8%, whereas random loss gave 51.7% and 62.9%, respectively. The loss-pattern ordering therefore depended on the classifier in this subset; the pilot does not establish a universal ordering.

After clean-validation calibration, mean negative log-likelihood increased from 0.162 to 3.636 for logistic regression and from 0.404 to 1.457 for

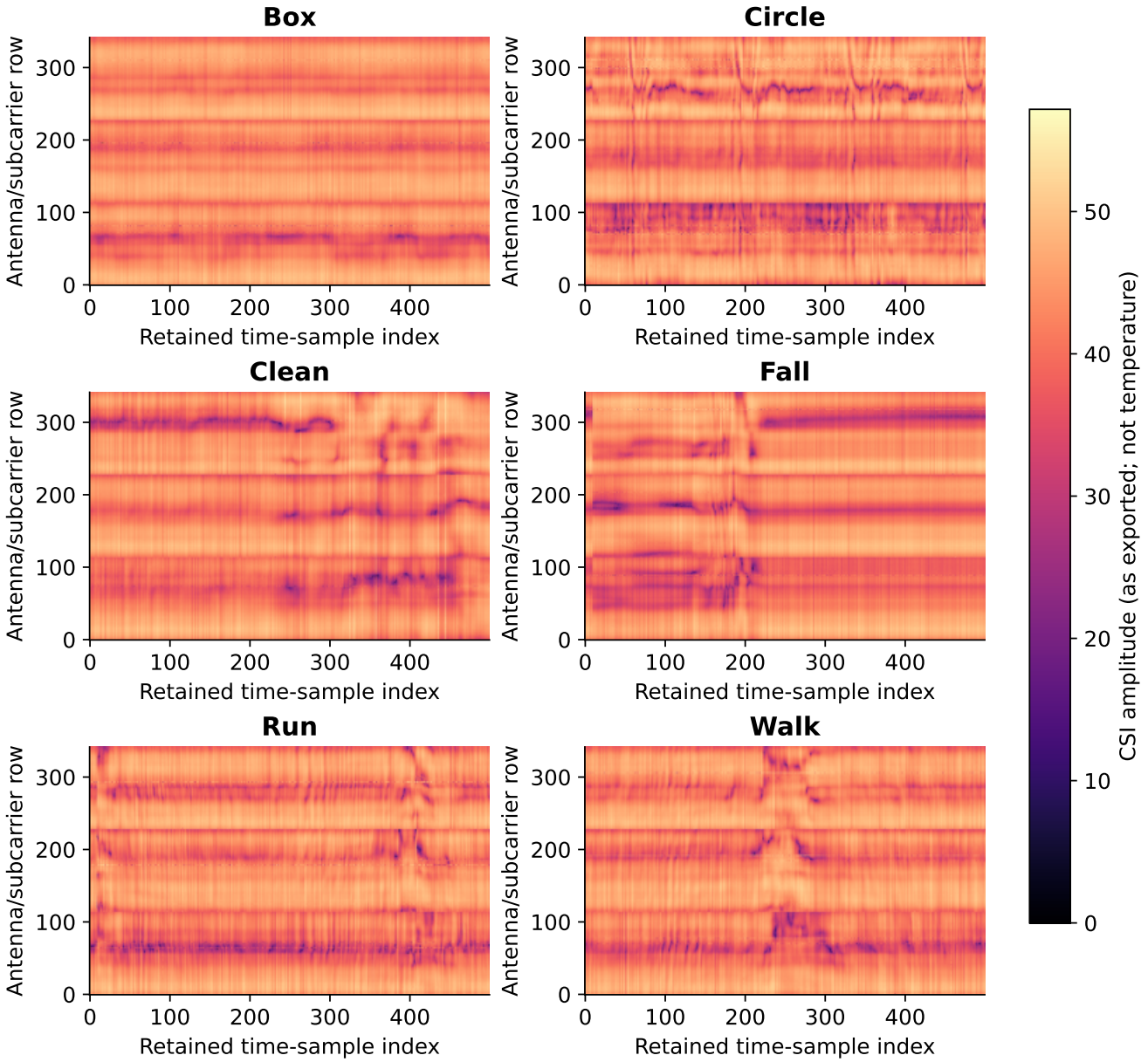


Figure 14. Genuine CSI amplitude examples for six activities. Each panel uses the lexically first selected training file of that class, with every fourth time sample retained. All panels share a color scale in the original exported amplitude units, without interpreting it as calibrated power or temperature. The horizontal coordinate is sample index, not elapsed seconds. The vertical coordinate stacks three antenna links and their 114 subcarriers.

the forest under 50% contiguous loss. Thus, equal accuracy can coexist with different probability errors. Risk–coverage values and per-file predictions are retained for further analysis.

These are descriptive results for one small, fixed subset. Five masks are repeated corruptions of the same files, not five independent datasets. Unavailable participant and session identifiers prevent stronger independence checks. The main transfer experiment uses a separately documented dataset; the pilot scores are not cross-environment results and are not pooled with that experiment.

Figure 14 contains genuine six-class NTU-Fi pilot inputs on their original exported amplitude scale. These examples, the 48-file pilot scores and its unavailable participant metadata remain separate from the seven-label, participant-disjoint main study. The full pilot manifests and saved predictions remain in the supplementary reproduction archive.

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